Cloud Security Operations & Incident Response

Centralised Logging · Flow Logs · Automated Alerting · Threat Detection · Dashboards · IR Playbook

Prepared by Sabitha S Karanavar · Azure subscription "Azure subscription 1" · October 2026 · Working draft

1. Scope and AWS → Azure mapping

The rubric names AWS services (CloudTrail, VPC Flow Logs, GuardDuty). HVMC runs on Microsoft Azure, so each requirement is met with the native Azure equivalent, deployed into the existing resource group `HVMC-CloudSecurity-RG` (East US) and the HVMC-Prod-VNet three-tier network.

Rubric / AWS service	HVMC Azure implementation	What it records or does	Coverage
AWS CloudTrail (management events)	Azure Activity Log -> diagnostic setting HVMC-ActivityLog-to-LAW	Every control-plane write/delete/action in the subscription: who, what, when, from which IP, result	8 categories
AWS CloudTrail (data events)	Resource logs: StorageRead/Write/Delete on hvmcehrstorage	Every read, write and delete of an ePHI blob, with caller identity and IP (HIPAA 164.312(b))	ePHI store
CloudTrail (session audit)	BastionAuditLogs on HVMC-Bastion	Who opened an admin session to which host, and when	Admin access
VPC Flow Logs	VNet flow log HVMC-Prod-VNet-FlowLog + Traffic Analytics	Allowed/denied 5-tuple flows for every NIC in the VNet, 30-day retention, analysed every 10 min	4 subnets
Security-group logs	NetworkSecurityGroupEvent / RuleCounter on the 3 NSGs	Which NSG rule matched how often	3 NSGs
CloudWatch Logs	Log Analytics workspace HVMC-SecOps-LAW	Single queryable store (KQL), 30-day retention, 0.5 GB/day cap	Central
CloudWatch Alarms / EventBridge + SNS	9 activity-log alert rules -> action group HVMC-SecOps-AG (e-mail)	Near-real-time notification of security-relevant changes	9 rules
AWS Budgets	Consumption budget HVMC-Lab-Budget ($10/month, 50/80/100 %)	Cost anomaly warning	Cost guard

2. Step 1 — Centralised logging, flow logs and automated alerting

The whole step is a single, re-runnable script (`scripts/01_setup_logging_alerting.sh`) so the configuration is documented as code and can be rebuilt or removed (`99_teardown.sh`). It was run on 15 October 2026 against the live subscription. Figure 1 shows the run.

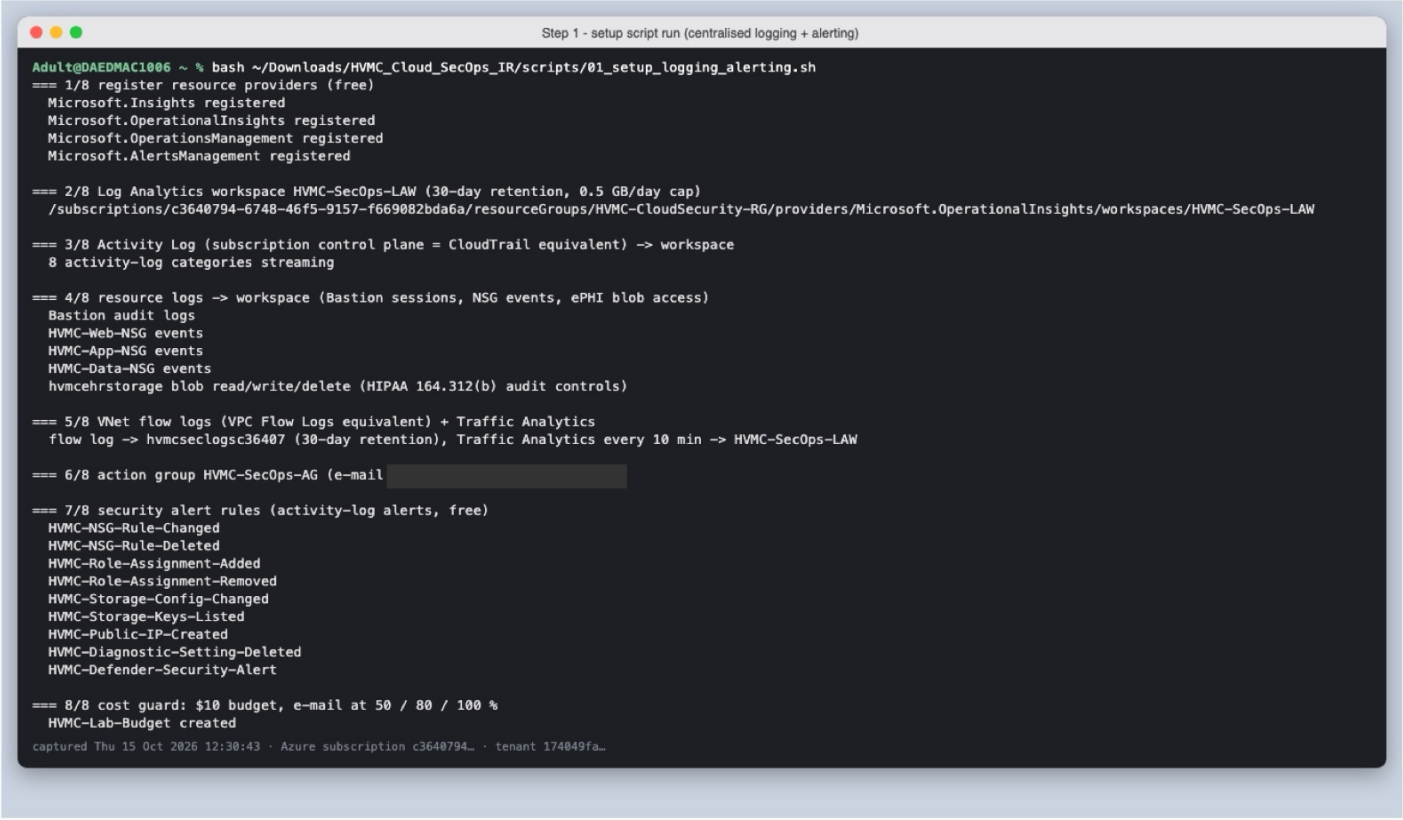


Figure 1. Setup script run: providers registered, workspace created, logs connected, flow logs, action group, 9 alert rules and budget.

2.1 Central log store

All security telemetry lands in one Log Analytics workspace, so a single KQL query can join who changed a firewall rule (Activity Log) with what traffic followed (flow logs). Retention is 30 days (inside the free retention period) and a 0.5 GB/day ingestion cap protects the free-trial credit.

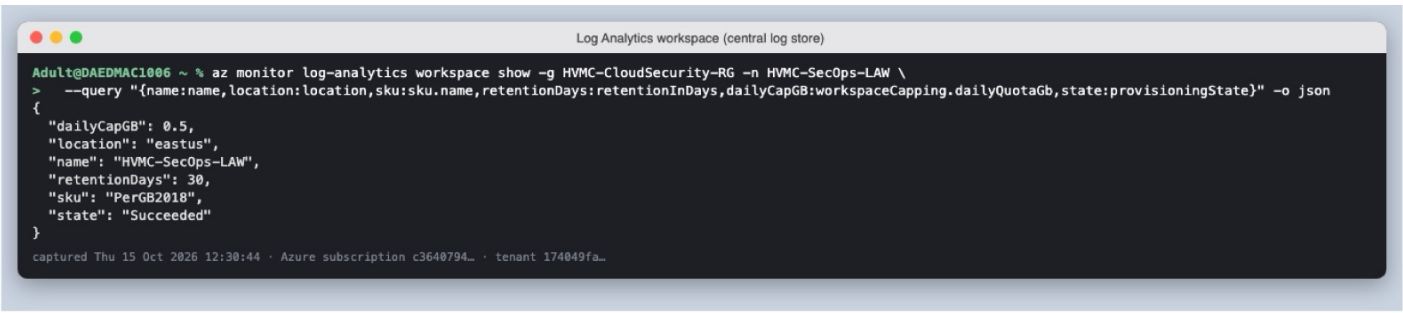


Figure 2. Log Analytics workspace HVMC-SecOps-LAW: PerGB2018, 30-day retention, 0.5 GB/day cap.

2.2 CloudTrail equivalent — Activity Log and resource audit logs

The subscription Activity Log is exported with all eight categories (Administrative, Security, ServiceHealth, Alert, Recommendation, Policy, Autoscale, ResourceHealth). Resource logs add the data-plane detail that the Activity Log does not hold: Bastion session audit, NSG rule events, and every read/write/delete on the ePHI storage account — the HIPAA §164.312(b) audit-control requirement that was flagged as a gap in the HIPAA analysis. Together with the alert rules below, this step treats risk **R01** in the HVMC risk register — “Security incident goes undetected for months (no central logging or alerting)”, the top-rated risk — whose finding was 0 workspaces, 0 activity-log exports, 0 alert rules and 0 flow logs.

Activity Log export - CloudTrail equivalent

```
Adult@DAEDMAC1006 ~ % az monitor diagnostic-settings subscription list \
>   --query "value[].{name:name,workspace:workspaceId,categories:logs[?enabled].category}" -o json
[
  {
    "categories": [
      "Administrative",
      "Security",
      "ServiceHealth",
      "Alert",
      "Recommendation",
      "Policy",
      "Autoscale",
      "ResourceHealth"
    ],
    "name": "HVMC-ActivityLog-to-LAW",
    "workspace": "/subscriptions/c3640794-6748-46f5-9157-f669082bda6a/resourceGroups/HVMC-CloudSecurity-RG/providers/Microsoft.OperationalInsights/workspaces/HVMC-SecOps-LAW"
  }
]
captured Thu 15 Oct 2026 12:30:46 · Azure subscription c3640794… · tenant 174049fa…
```

Figure 3. Subscription Activity Log exported to the workspace with all eight categories.

Resource logs - Bastion, NSGs, ePHI storage

```
Adult@DAEDMAC1006 ~ % Q="[].[name,join(',',logs[?enabled].category)]"
Adult@DAEDMAC1006 ~ % for n in HVMC-Web-NSG HVMC-App-NSG HVMC-Data-NSG; do printf '%-22s' $n; az monitor diagnostic-settings list --resource $(az network nsg show -g HVMC-CloudSecurity-RG -n $n --query id -o tsv) --query "$Q" -o tsv; done
Adult@DAEDMAC1006 ~ % printf '%-22s' HVMC-Bastion; az monitor diagnostic-settings list --resource $(az network bastion show -g HVMC-CloudSecurity-RG -n HVMC-Bastion --query id -o tsv) --query "$Q" -o tsv
Adult@DAEDMAC1006 ~ % printf '%-22s' hvmcehrstorage/blob; az monitor diagnostic-settings list --resource $(az storage account show -g HVMC-CloudSecurity-RG -n hvmcehrstorage --query id -o tsv)/blobServices/default --query "$Q" -o tsv
HVMC-Web-NSG           HVMC-to-LAW       NetworkSecurityGroupEvent,NetworkSecurityGroupRuleCounter
HVMC-App-NSG           HVMC-to-LAW       NetworkSecurityGroupEvent,NetworkSecurityGroupRuleCounter
HVMC-Data-NSG          HVMC-to-LAW       NetworkSecurityGroupEvent,NetworkSecurityGroupRuleCounter
HVMC-Bastion           HVMC-to-LAW       BastionAuditLogs
hvmcehrstorage/blob    HVMC-ePHI-access-to-LAW   StorageRead,StorageWrite,StorageDelete
captured Thu 15 Oct 2026 12:30:59 · Azure subscription c3640794… · tenant 174049fa…
```

Figure 4. Resource logs: NSG events on all three tiers, Bastion audit logs, and ePHI blob access on hvmcehrstorage.

2.3 VPC Flow Logs equivalent — VNet flow logs

A VNet-level flow log (the successor to NSG flow logs, which Azure stopped accepting for new deployments in 2025) records every flow through HVMC-Prod-VNet into a dedicated log storage account kept separate from the ePHI account, with 30-day retention. Traffic Analytics processes the flows every 10 minutes into the workspace. *Note:* no virtual machines run in the VNet, so flow volume is low (Bastion and load-balancer traffic only); by 14:35 local time on 15 October Traffic Analytics had already written 99 flow records (NTANetAnalytics) and 139 IP-detail records to the workspace (see the Workbook figure in section 4).

VNet flow logs + Traffic Analytics - VPC Flow Logs equivalent

```
Adult@DAEDMAC1006 ~ % az network watcher flow-log show --location eastus -n HVMC-Prod-VNet-FlowLog \
>   --query "{name:name,target:targetResourceId,enabled:enabled,storage:storageId,retentionDays:retentionPolicy.days,trafficAnalytics:flowAnalyticsConfiguration.networkWatcherFlowAnalyticsConfiguration.enabled,intervalMin:flowAnalyticsConfiguration.networkWatcherFlowAnalyticsConfiguration.trafficAnalyticsInterval}" -o json
{
  "enabled": true,
  "intervalMin": 10,
  "name": "HVMC-Prod-VNet-FlowLog",
  "retentionDays": 30,
  "storage": "/subscriptions/c3640794-6748-46f5-9157-f669082bda6a/resourceGroups/HVMC-CloudSecurity-RG/providers/Microsoft.Storage/storageAccounts/hvmcseclogsc36407",
  "target": "/subscriptions/c3640794-6748-46f5-9157-f669082bda6a/resourceGroups/HVMC-CloudSecurity-RG/providers/Microsoft.Network/virtualNetworks/HVMC-Prod-VNet",
  "trafficAnalytics": true
}
captured Thu 15 Oct 2026 12:31:01 · Azure subscription c3640794… · tenant 174049fa…
```

Figure 5. VNet flow log on HVMC-Prod-VNet with 30-day retention and Traffic Analytics every 10 minutes.

2.4 Automated alerting

Nine activity-log alert rules watch for the changes that matter most for HVMC's risk profile. Each rule is mapped to a risk in the register, an incident response playbook and a MITRE ATT&CK technique, and notifies the Security Lead by e-mail through the action group HVMC-SecOps-AG (common alert schema).

Alert rule	Trigger (operation)	Risk	Playbook	Threat / ATT&CK
HVMC-NSG-Rule-Changed	`securityRules/write`	R03	PB-01 Internet exposure	Initial access (T1190 enabler)
HVMC-NSG-Rule-Deleted	`securityRules/delete`	R03	PB-01	Defence evasion
HVMC-Role-Assignment-Added	`roleAssignments/write`	R04	PB-02 Privilege escalation	T1098 Account manipulation
HVMC-Role-Assignment-Removed	`roleAssignments/delete`	R04	PB-02	Impact / cover tracks
HVMC-Storage-Config-Changed	`storageAccounts/write`	R02	PB-03 ePHI exposure	T1530 Data from cloud storage
HVMC-Storage-Keys-Listed	`storageAccounts/listKeys/action`	R02, R05	PB-03	T1552 Unsecured credentials
HVMC-Public-IP-Created	`publicIPAddresses/write`	R03, R10	PB-01	New attack surface
HVMC-Diagnostic-Setting-Deleted	`diagnosticSettings/delete`	R01	PB-04 Logging tampering	T1562.008 Disable cloud logs
HVMC-Defender-Security-Alert	`category = Security`	R11, all	per alert type	Any Defender detection

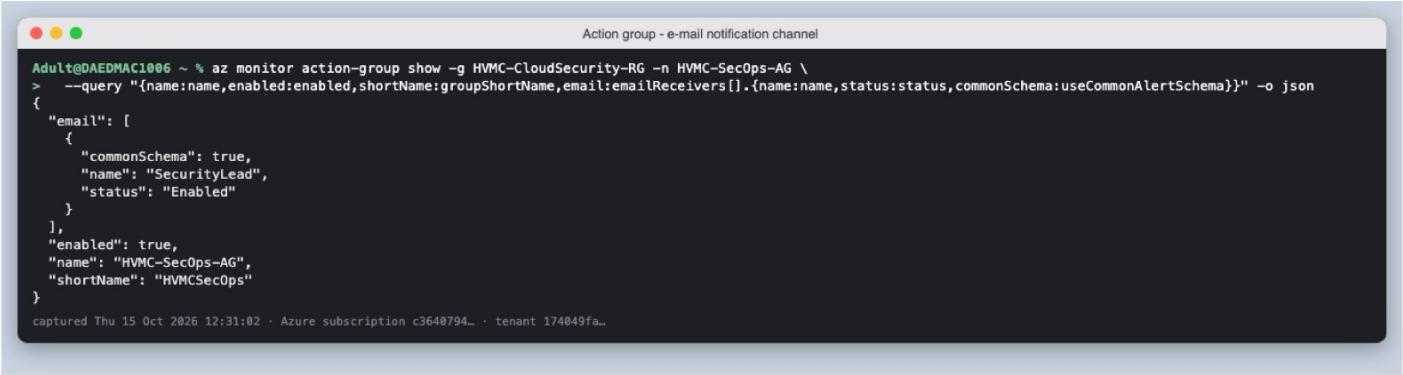


Figure 6. Action group HVMC-SecOps-AG: e-mail receiver for the Security Lead.

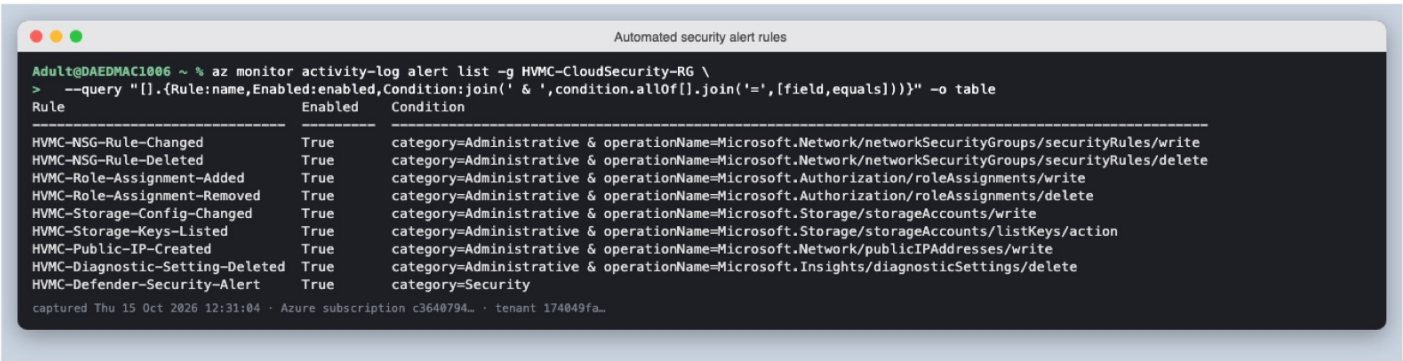


Figure 7. The nine automated security alert rules, all enabled.

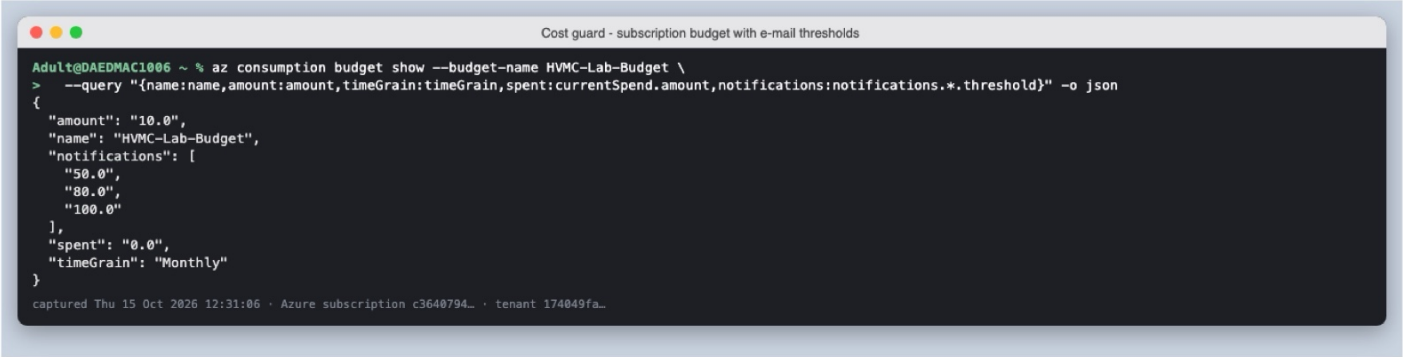


Figure 8. Cost guard: $10 monthly budget with e-mail warnings at 50, 80 and 100 percent.

3. Step 2 — Threat detection (GuardDuty equivalent)

Logging and rule-based alerts (Step 1) only catch what was anticipated. Step 2 adds two layers that look for *suspicious behaviour*, the role GuardDuty plays in AWS, deployed with `scripts/02_setup_threat_detection.sh`:

Layer	Service	What it detects for HVMC	Cost
Cloud-native threat detection	Microsoft Defender for Storage (Defender for Storage V2)	Unusual access to the ePHI account (access from Tor / unusual location, anomalous data extraction, suspicious SAS use) and **on-upload malware scanning** of every new blob, with the result written as a blob index tag. Risks R02, R05, **R11**.	30-day free trial; scanning capped at 1 GB/month/account ($0.15/GB)
Control-plane threat detection	Microsoft Defender for Resource Manager	Suspicious management operations: known attacker toolkits (MicroBurst, PowerZure), access from malicious IPs, unusual permission or resource changes. Risks R04, R08.	30-day free trial
SIEM / correlation	Microsoft Sentinel on HVMC-SecOps-LAW	Correlates the Step 1 logs and the Defender alerts; analytics rules turn detections into tracked **incidents** with owner, status and entities (account, IP, resource).	31-day free trial (10 GB/day)
Notification	Defender security contact	E-mails the Security Lead for every Defender alert of Medium severity or higher.	Free

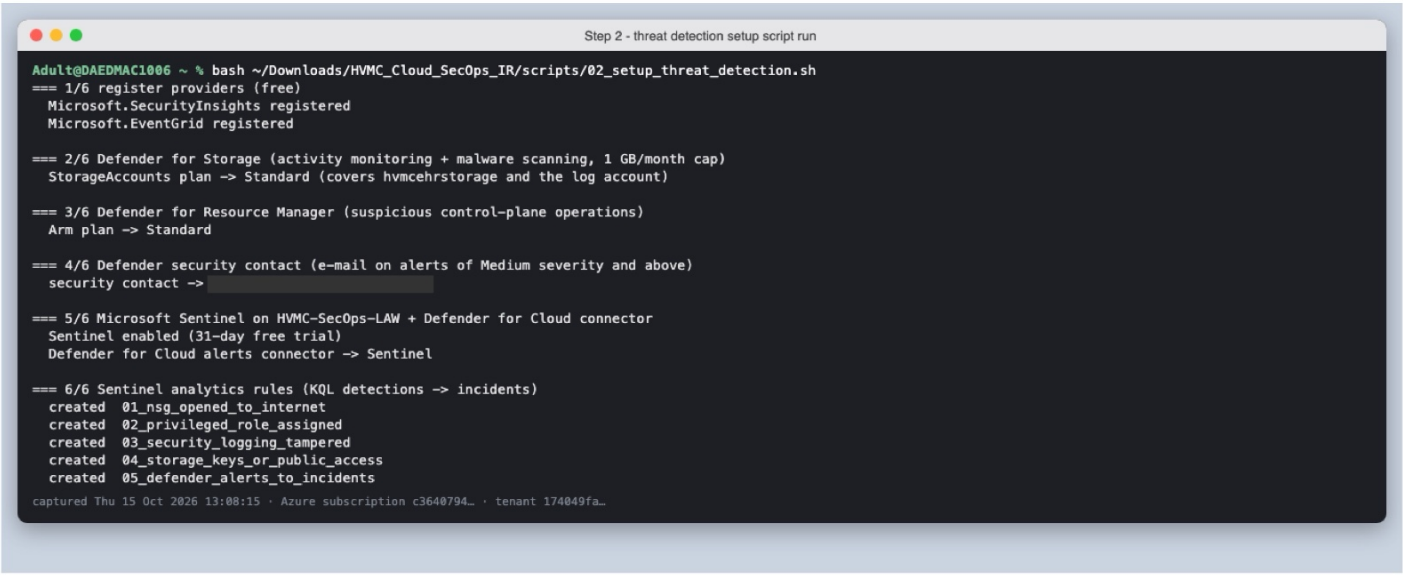


Figure 9. Step 2 setup script run: Defender plans, security contact, Sentinel, connector and five analytics rules created.

Microsoft Defender for Cloud plans - GuardDuty equivalent

```
Adult@DAEDMAC1006 ~ % az security pricing list \
>   --query "value[?contains(['StorageAccounts','Arm','CloudPosture','FoundationalCspm'],name)].{Plan:name,Tier:pricingTier,SubPlan:subPlan,FreeTrialLeft:freeTrialR
emainingTime}" -o table
Plan               Tier      SubPlan                FreeTrialLeft
-----------------  --------  ---------------------  -----------------
StorageAccounts    Standard  DefenderForStorageV2   29 days, 23:58:00
Arm                Standard  PerSubscription        29 days, 23:58:00
CloudPosture       Free                             30 days, 0:00:00
FoundationalCspm   Standard                         0:00:00
captured Thu 15 Oct 2026 13:08:17 · Azure subscription c3640794… · tenant 174049fa…
```

Figure 10. Defender for Cloud: Storage and Resource Manager plans on Standard with the 30-day free trial running.

Defender for Storage plan - activity monitoring + on-upload malware scanning (all storage incl. ePHI)

```
Adult@DAEDMAC1006 ~ % az rest --method get --url "https://management.azure.com/subscriptions/c3640794-6748-46f5-9157-f669082bda6a/providers/Microsoft.Security/prici
ngs/StorageAccounts?api-version=2024-01-01" \
>   --query "{plan:name,tier:properties.pricingTier,subPlan:properties.subPlan,trialLeft:properties.freeTrialRemainingTime,extensions:properties.extensions[].{name:
name,enabled:isEnabled,settings:additionalExtensionProperties}}" -o json
Adult@DAEDMAC1006 ~ % az storage account list -g HVMC-CloudSecurity-RG --query "[].{ProtectedStorageAccount:name}" -o table
{
  "extensions": [
    {
      "enabled": "True",
      "name": "OnUploadMalwareScanning",
      "settings": {
        "AutomatedResponse": "None",
        "BlobScanResultsOptions": "BlobIndexTags",
        "CapGBPerMonthPerStorageAccount": "1"
      }
    },
    {
      "enabled": "False",
      "name": "SensitiveDataDiscovery",
      "settings": null
    }
  ],
  "plan": "StorageAccounts",
  "subPlan": "DefenderForStorageV2",
  "tier": "Standard",
  "trialLeft": "P29DT23H58M"
}
ProtectedStorageAccount
------------------------
hvmcehrstorage
hvmcseclogsc36407
captured Thu 15 Oct 2026 13:08:19 · Azure subscription c3640794… · tenant 174049fa…
```

Figure 11. Defender for Storage plan settings: malware scanning on upload (1 GB/month cap) protecting both storage accounts, including the ePHI store.

Defender security contact - alert e-mail notifications

```
Adult@DAEDMAC1006 ~ % az rest --method get --url "https://management.azure.com/subscriptions/c3640794-6748-46f5-9157-f669082bda6a/providers/Microsoft.Security/securi
tyContacts/default?api-version=2023-12-01-preview" \
>   --query "{emails:properties.emails,enabled:properties.isEnabled,notifyRoles:properties.notificationsByRole.roles,alertMinSeverity:properties.notificationsSource
s[0].minimalSeverity}" -o json
{
  "alertMinSeverity": "Medium",
  "emails":
  "enabled": true,
  "notifyRoles": [
    "Owner"
  ]
}
captured Thu 15 Oct 2026 13:08:21 · Azure subscription c3640794… · tenant 174049fa…
```

Figure 12. Defender security contact: e-mail to the Security Lead and subscription Owners for Medium+ alerts.

3.1 Sentinel analytics rules

Four custom scheduled KQL rules run every five minutes over a 15-minute window, and a Microsoft-security rule turns every Defender alert into an incident. Sentinel also enables its built-in **Fusion** rule, which correlates low-fidelity signals into multistage-attack incidents. The custom rules go further than the Step 1 alerts: they read the request body, so a firewall change only becomes a High incident when it really *allows inbound traffic from the Internet*, and a role assignment only when the role is Owner, Contributor or User Access Administrator. Each query was tested against the live workspace before deployment, and each rule is stored as JSON in `scripts/sentinel_rules/`.

Rule	Severity	MITRE ATT&CK	Risk	Playbook
NSG inbound rule opened to the Internet	High	Initial Access, Defense Evasion (T1190, T1562)	R03, R10	PB-01
Privileged Azure role assigned	High	Privilege Escalation, Persistence (T1098)	R04, R07	PB-02
Security logging disabled or deleted (incl. Defender plan downgraded)	High	Defense Evasion (T1562)	R01	PB-04
ePHI storage keys retrieved or storage exposure changed	Medium	Credential Access, Collection (T1552, T1530)	R02, R05	PB-03
Incidents from Defender for Cloud alerts	per alert	per alert	R11 and others	by alert type

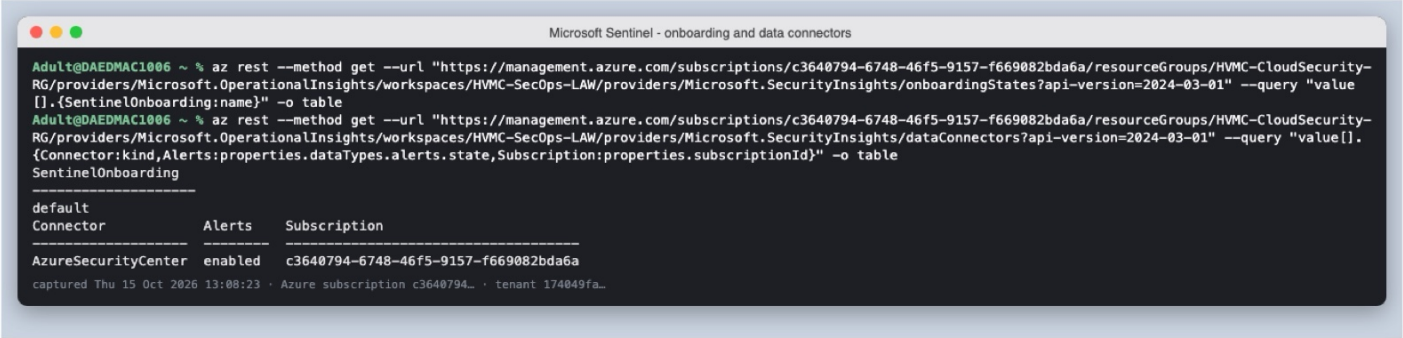


Figure 13. Microsoft Sentinel onboarded on HVMC-SecOps-LAW with the Defender for Cloud alerts connector enabled.

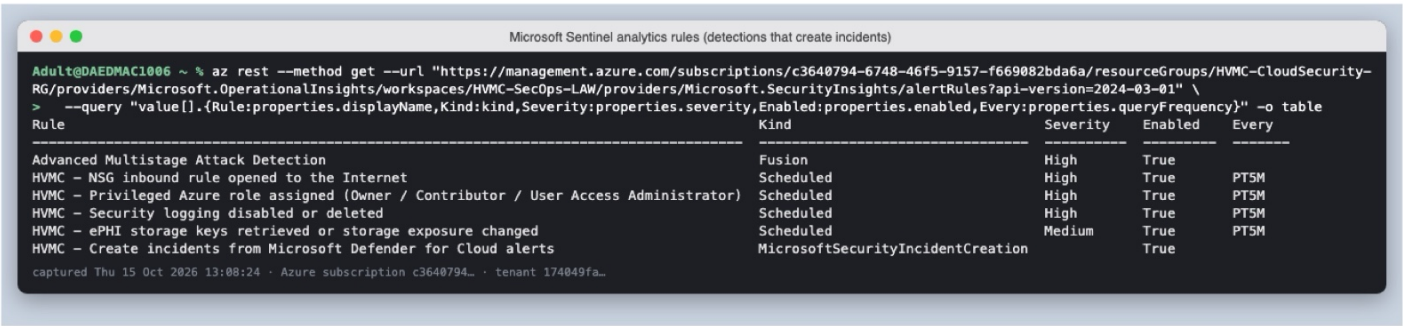


Figure 14. Sentinel analytics rules: four HVMC scheduled detections, the Defender incident rule and the built-in Fusion rule.

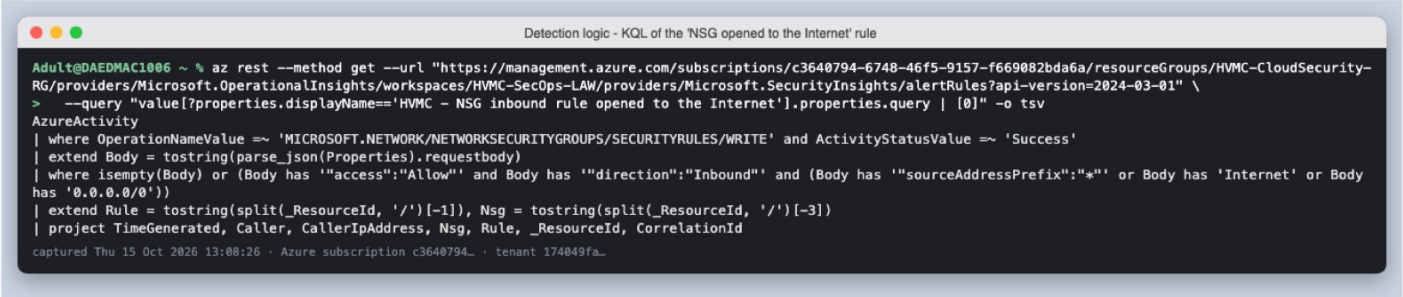


Figure 15. Detection logic of the 'NSG inbound rule opened to the Internet' rule (KQL).

Not included: Microsoft Entra ID sign-in and audit logs. Exporting them to a workspace requires an Entra ID P1/P2 licence, which the lab tenant does not have; they remain viewable in the Entra portal for 7 days. This is recorded as a known gap (risk R12, AiTM phishing).

4. Step 3 — Security dashboards and key metrics

Two dashboards present the same telemetry for different audiences. Both are free: they only read data that is already collected.

Dashboard	Audience	Built with	How it is refreshed
HVMC Security Operations Dashboard (HTML)	Security Lead, IT Manager, auditors — one page, no Azure login needed	`dashboard/collect_metrics.py` (read-only Azure CLI + KQL) → `build_dashboard.py` ; triage notes in `triage.json`	Re-run the two scripts (same pattern as the KRI monitor from the risk work); light and dark mode, hover tooltips
HVMC Security Operations Azure Workbook	SOC analyst working inside Azure / Sentinel	`scripts/03_create_workbook.sh` , 10 KQL tiles with a time-range picker (`dashboard/workbook_queries.py`)	Live: each tile queries HVMC-SecOps-LAW when opened

4.1 Key metrics

Metric	Definition	Source	Target / threshold	Value 15 Oct 2026, after the exercise
Log source coverage	Security log sources connected ÷ sources required	Diagnostic settings, flow log, Defender plans, Sentinel	100 %; any “Missing” = Red	10 / 11 (Entra ID logs: licence gap)
Detection rules enabled	Enabled alert + analytics rules	Azure Monitor, Sentinel	All enabled; a disabled rule = Red (PB-04)	15 / 15
Alerts fired / untriaged	Alerts in 7 days; alerts without a disposition	Azure Monitor alerts + triage log	Untriaged = 0 within 1 business day	40 fired · 0 untriaged
Open incidents · MTTA · MTTR	Open Sentinel incidents; mean minutes created → first action → closed	SecurityIncident	High: acknowledge ≤ 15 min, contain ≤ 1 h	0 open · 8 closed (4 duplicates after grouping fix)
Security-relevant changes	Firewall, role, storage, key, public-IP, logging and Defender changes	AzureActivity	Each must match an approved change (IP-1)	10 (all from approved changes)
Defender secure score	Defender for Cloud secure score	Defender for Cloud	≥ 80 % by Phase 2 of the risk program	50 % (6 / 12)
Ingestion vs cap	MB ingested per day vs 0.5 GB cap	Usage	< 80 % of cap	< 1 MB/day
Credit used	Trial credit consumed, by resource type	Cost Management	SecOps tooling = $0	$53.35 total · SecOps $0.00

4.2 First real detections and triage

The alerting worked before any test was run. Eleven alerts fired during setup and were triaged within the hour; both groups were authorised changes, so they are recorded as benign with evidence (the dashboard below also shows the 29 alerts from the Step 4 exercise):

- **16:32 UTC — HVMC-Storage-Config-Changed (×3):** the Step 1 script created the flow-log storage account. Caller verified (Sabitha S Karanavar via Azure CLI); the ePHI account was not touched.
- **17:12 UTC — HVMC-Role-Assignment-Added (×8):** enabling Defender for Storage and Sentinel caused Azure to grant roles to Microsoft service principals — *Defender for Storage Scanner Operator* and *EventGrid Contributor* at subscription scope, and *User Access Administrator* scoped only to the Sentinel workspace. These are expected platform identities; recording them stops them being mistaken for privilege escalation later (lesson learned: pre-register planned changes so their alerts can be auto-closed).

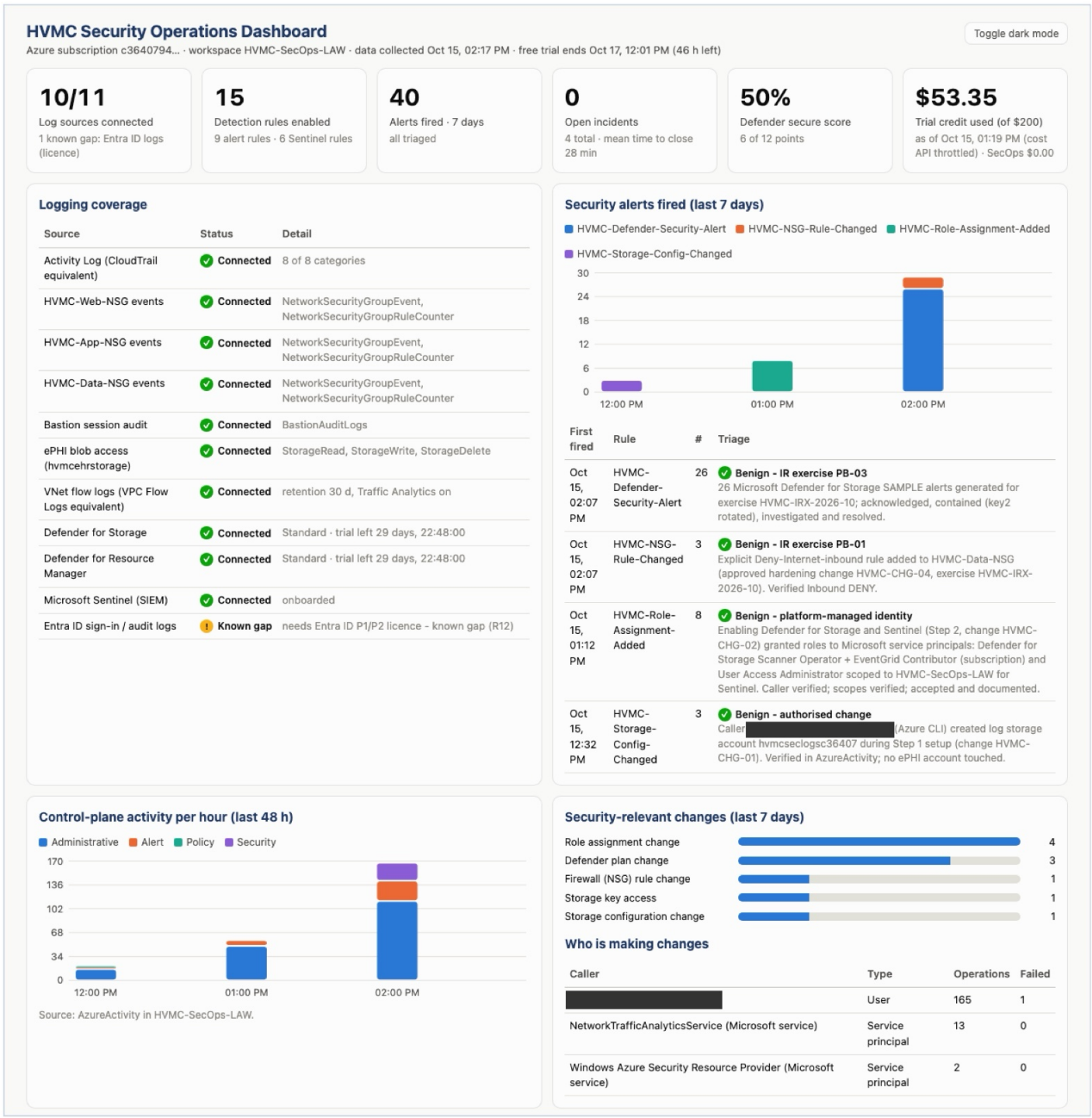


Figure 16. HVMC Security Operations Dashboard (top): KPI tiles, logging coverage, alerts fired with triage dispositions, activity and changes.

Incidents (Microsoft Sentinel)

#	Title	Severity	Status	Created	Time to close
1	HVMC - Privileged Azure role assigned (Owner / Contributor / User Access Administrator)	High	Closed	Oct 15, 01:17 PM	57 min
4	HVMC - NSG rule change on hvmc-data-nsg: rule body not logged - verify access and direction	Medium	Closed	Oct 15, 02:15 PM	0 min
3	HVMC - NSG inbound rule opened to the Internet	High	Closed	Oct 15, 02:12 PM	2 min
2	HVMC - Privileged Azure role assigned (Owner / Contributor / User Access Administrator)	High	Closed	Oct 15, 01:22 PM	52 min

Detection rules

Rule	Type	Severity	Status
HVMC-NSG-Rule-Changed	Activity-log alert	—	Enabled
HVMC-NSG-Rule-Deleted	Activity-log alert	—	Enabled
HVMC-Role-Assignment-Added	Activity-log alert	—	Enabled
HVMC-Role-Assignment-Removed	Activity-log alert	—	Enabled
HVMC-Storage-Config-Changed	Activity-log alert	—	Enabled
HVMC-Storage-Keys-Listed	Activity-log alert	—	Enabled
HVMC-Public-IP-Created	Activity-log alert	—	Enabled
HVMC-Diagnostic-Setting-Deleted	Activity-log alert	—	Enabled
HVMC-Defender-Security-Alert	Activity-log alert	—	Enabled
Advanced Multistage Attack Detection	Sentinel Fusion	High	Enabled
HVMC - NSG inbound rule opened to the Internet	Sentinel Scheduled	High	Enabled
HVMC - Privileged Azure role assigned (Owner / Contributor / User Access Administrator)	Sentinel Scheduled	High	Enabled
HVMC - Security logging disabled or deleted	Sentinel Scheduled	High	Enabled
HVMC - ePHI storage keys retrieved or storage exposure changed	Sentinel Scheduled	Medium	Enabled
HVMC - Create incidents from Microsoft Defender for Cloud alerts	Sentinel MicrosoftSecurityIncidentCreation	per alert	Enabled

Security posture (Defender for Cloud)

50% secure score · 6 / 12

5 healthy · 10 unhealthy · 13 not applicable. Defender re-evaluates about every 24 h, so items fixed in Step 2 may still be listed.

Open recommendation	Status
Email notification to subscription owner for high severity alerts should be enabled	Unhealthy
Microsoft Defender for Resource Manager should be enabled	Unhealthy
Email notification for high severity alerts should be enabled	Unhealthy
Subscriptions should have a contact email address for security issues	Unhealthy
Microsoft Defender CSPM should be enabled	Unhealthy
Storage account should use a private link connection	Unhealthy
Storage accounts should restrict network access using virtual network rules	Unhealthy
Storage accounts should prevent shared key access	Unhealthy
Microsoft Defender for Storage plan should be enabled with Malware Scanning and Sensitive Data Threat Detection	Unhealthy
There should be more than one owner assigned to subscriptions	Unhealthy

Cost guard (trial credit used)

$53.35 of $200 free-trial credit · spending limit on, no card charge

bastionhosts	$39.58
natgateways	$10.41
publicipaddresses	$3.36
loadbalancers	$0.00
storageaccounts	$0.00

All credit use comes from the pre-existing Bastion, NAT gateway and public IPs. Security-operations tooling (workspace, alerts, Defender and Sentinel trials) has cost $0.00 so far. Cost data can lag by up to 24 h.

Read-only data collected by collect_metrics.py · triage dispositions from triage.json · Horizon Valley Medical Center · Sabitha S Karanavar

Figure 17. HVMC Security Operations Dashboard (bottom): incidents, detection rules, Defender posture and cost guard.

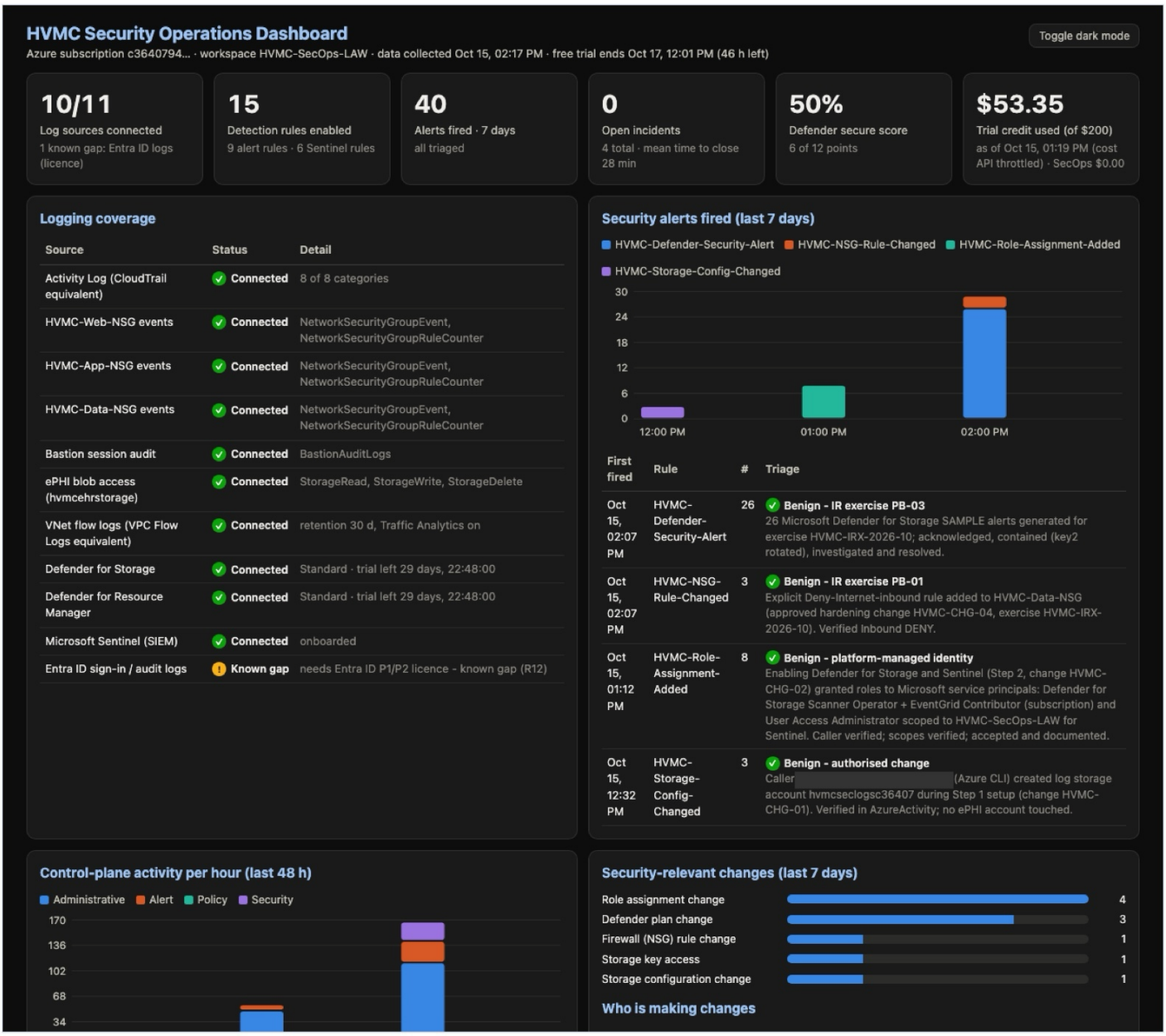


Figure 18. The same dashboard in dark mode (each colour step re-chosen for the dark surface, not inverted).

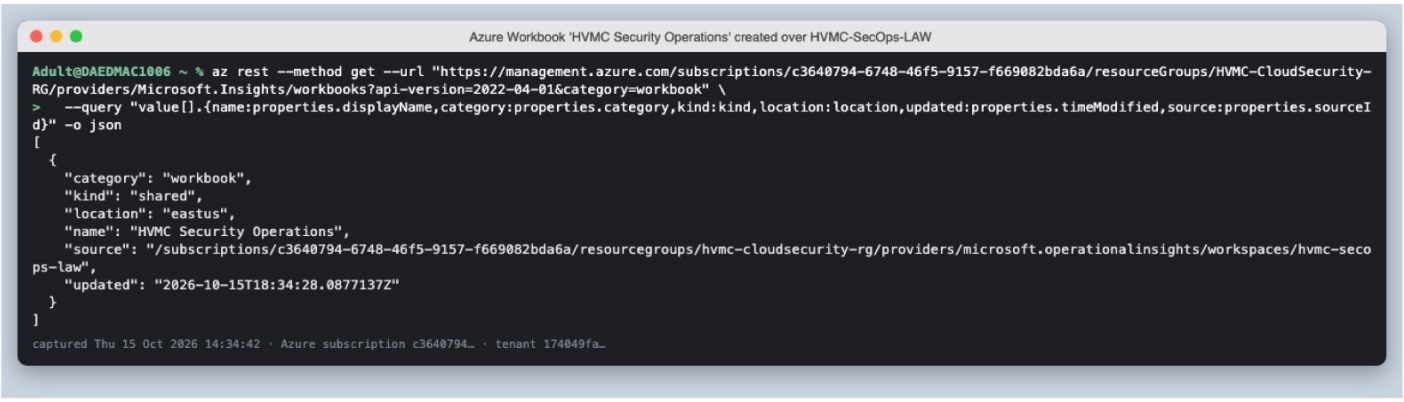


Figure 19. Azure Workbook 'HVMC Security Operations' deployed by scripts/03_create_workbook.sh, linked to HVMC-SecOps-LAW.

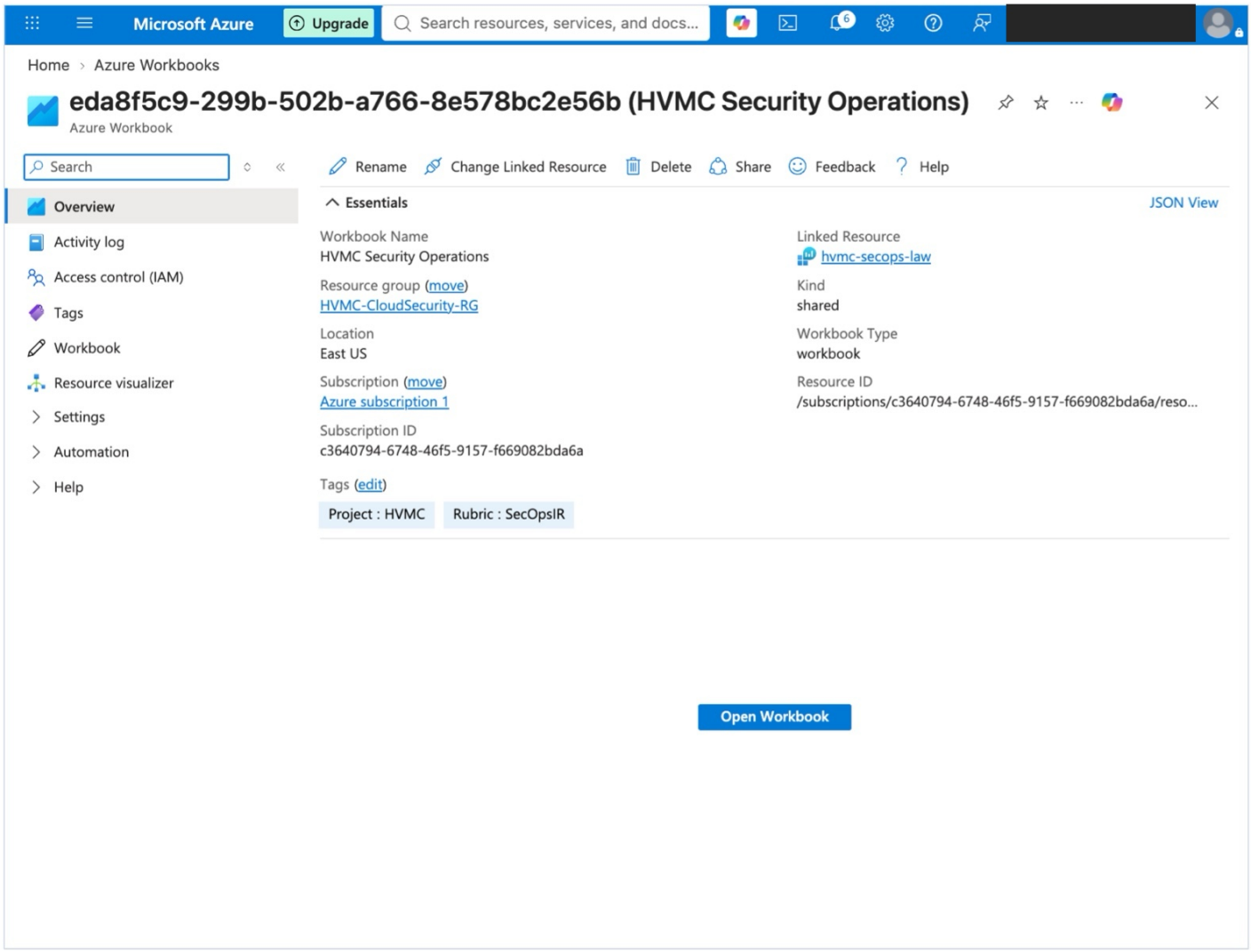


Figure 20. The workbook resource in the Azure portal: linked to hvmc-secops-law, HVMC-CloudSecurity-RG, tagged Project HVMC / Rubric SecOpsIR.

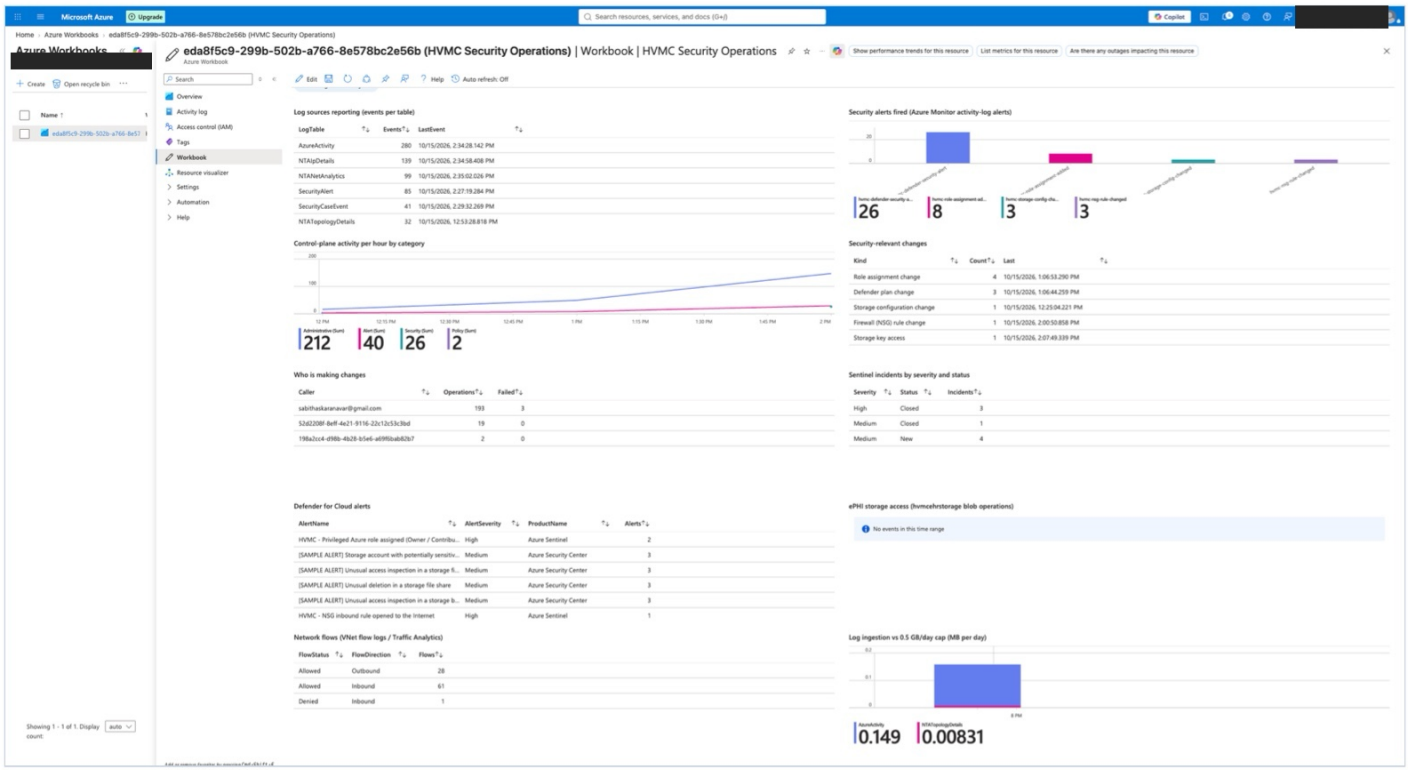


Figure 21. Azure Workbook 'HVMC Security Operations' in the portal, all ten tiles (Last 7 days): log sources, alerts per rule, activity per hour, security changes, callers, Sentinel incidents, Defender alerts, ePHI access, network flows (61 inbound / 28 outbound allowed, 1 denied) and ingestion vs cap. Captured before the duplicate-incident triage, so 4 Medium incidents still show as New (closed at 18:44 UTC - see section 5).

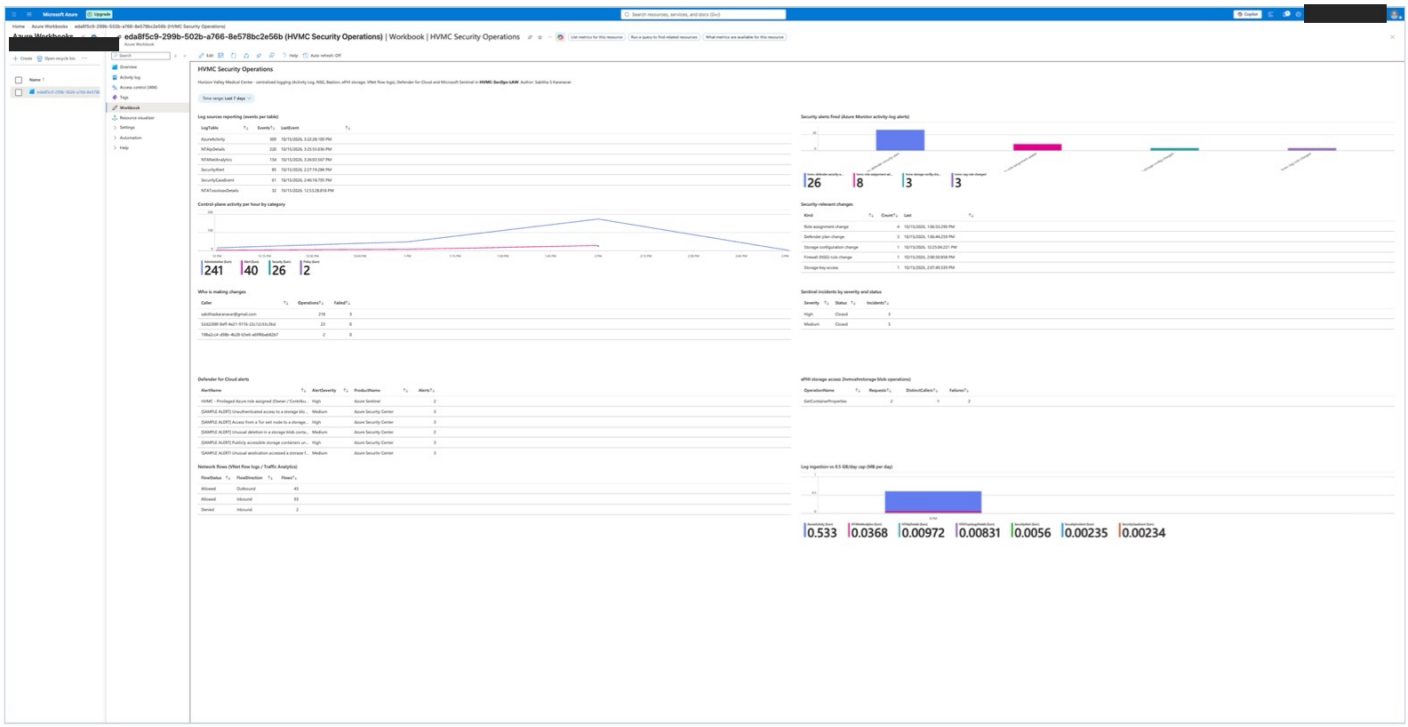


Figure 22. The same Workbook after triage: Sentinel incidents High 3 Closed and Medium 5 Closed (none New); flows 93 inbound / 45 outbound allowed, 2 denied; first ePHI storage access records.

Triage of the first ePHI storage access records (Workbook, 19:11 UTC): two `GetContainerProperties` requests with status 404 on `hvmcehrstorage`. Investigation in StorageBlobLogs showed authentication type *TrustedAccess*, user agent `SRP/1.0` (the Azure Storage Resource Provider) checking for the Microsoft Purview system container

`$purview`, which does not exist. No blob was listed or read and no identity of HVMC staff was involved, so the activity was classified as benign platform behaviour. This confirms the ePHI audit log (HIPAA §164.312(b)) records even platform-level access.

5. Step 4 — Incident response playbook and testing

The full playbook is a separate document, **HVMC_Cloud_IR_Playbook.pdf** (version 1.0). It defines roles and a RACI, a four-level severity scale with response targets, HIPAA breach-notification steps, a common five-phase procedure (identification, containment, investigation, eradication, recovery) and four scenario playbooks, each tied to the detections built in Steps 1–2:

Playbook	Scenario	Detections that trigger it	Risks	How it was tested
PB-01	Internet exposure / unauthorised firewall change	HVMC-NSG-Rule-Changed/-Deleted, Public-IP-Created; Sentinel “NSG opened to the Internet”	R03, R10	Live test
PB-02	Privilege escalation / privileged role misuse	HVMC-Role-Assignment-Added; Sentinel “Privileged Azure role assigned”; Defender for Resource Manager	R04, R07	Tabletop exercise
PB-03	ePHI storage threat (malware, unusual access, key misuse)	Defender for Storage alerts; HVMC-Storage-Keys-Listed; Sentinel “ePHI storage keys retrieved”	R02, R05, R11	Live test
PB-04	Security logging disabled or tampered	HVMC-Diagnostic-Setting-Deleted; Sentinel “Security logging disabled”	R01	Scheduled (Phase 2)

Test design principle: no test weakened a security control. PB-03 used Microsoft Defender for Cloud's built-in sample alerts; PB-01 used a real, approved *hardening* change (an explicit Deny-Internet-inbound rule on the Data-tier NSG, which also treats risk R03); PB-02, whose live version would require granting excess privilege, was run as a tabletop exercise on real alert data.

5.1 Live test results (exercise HVMC-IRX-2026-10, 15 Oct 2026, times UTC)

Test	Triggered	Detected	Time to detect (min)	Detect → contain (min)	Detect → recover (min)	Phases passed
PB-03	18:03:31	18:03:32	0.0	4.3	4.7	6/6
PB-01	18:00:49	18:07:26	6.6	0.2	0.3	4/4

PB-03: ePHI storage threat - Defender for Storage alerts (built-in sample alerts)

Phase	Done	Result	Evidence
1 Identification	18:07:24	Pass	26 Defender alerts: [SAMPLE ALERT] Access from a Tor exit node to a storage blob container; [SAMPLE ALERT] Access from a Tor exit node to a storage file share; [SAMPLE ALERT] Access from a suspicious IP to a storage blob container; [SAMPLE ALERT] Access from a suspicious IP to a storage file share;
1b Triage / acknowledge	18:07:48	Pass	26 alerts set to In progress (acknowledged)
2 Containment	18:07:50	Pass	secondary access key of hvmcehrstorage regenerated (key2 not used by any application); network posture recorded: {'defaultAction': 'Allow', 'publicNetworkAccess': 'Enabled', 'sharedKey': True}
3 Investigation	18:07:52	Pass	alert entities reviewed; ePHI blob data-plane operations in 24 h: 0; control-plane events on the account in 24 h: 0 (all by known callers)

4 Eradication	18:07:52	Pass	Defender for Storage Standard and malware scanning True confirmed; sample alerts carry no payload, so no blob needed removal
5 Recovery	18:08:15	Pass	26 Defender alerts resolved; protection verified still enabled

PB-01: Unplanned firewall change on the Data tier - detection and change verification

Phase	Done	Result	Evidence
1 Identification	18:07:39	Pass	Azure Monitor alert HVMC-NSG-Rule-Changed fired at 2026-10-15T18:07:26Z; e-mail sent to the Security Lead
2-3 Triage, containment decision and investigation	18:07:41	Pass	who: [redacted] what: Inbound Deny from Internet to port *; matches approved change HVMC-CHG-04 -> authorised; effect reduces exposure
4 Eradication (exposure sweep)	18:07:44	Pass	sweep of all NSGs: no Internet-facing allow rule except the approved Web HTTPS 443 rule
5 Recovery (baseline updated)	18:07:45	Pass	baseline updated with the approved change; difference from previous baseline: ['Deny-Internet-Inbound-Data p4000 Deny Internet->*']; App->Data 1433 rule unchanged

Microsoft Sentinel incidents raised and closed

#	Title	Severity	Created	Closed	Minutes open	Classification
#1	HVMC - Privileged Azure role assigned (Owner / Contributor / User Access Administrator)	High	17:17:18	18:14:21	57.0	BenignPositive / SuspiciousButExpected
#2	HVMC - Privileged Azure role assigned (Owner / Contributor / User Access Administrator)	High	17:22:19	18:14:18	52.0	BenignPositive / SuspiciousButExpected
#3	HVMC - NSG inbound rule opened to the Internet	High	18:12:22	18:14:15	1.9	FalsePositive / IncorrectAlertLogic
#4	HVMC - NSG rule change on hvmc-data-nsg: rule body not logged - verify access and direction	Medium	18:15:05	18:15:39	0.6	BenignPositive / SuspiciousButExpected
#5	HVMC - ePHI storage keys retrieved or storage exposure changed	Medium	18:17:30	18:44:16	26.8	BenignPositive / SuspiciousButExpected
#6	HVMC - NSG rule change on hvmc-data-nsg: rule body not logged - verify access and direction	Medium	18:20:07	18:44:13	24.1	BenignPositive / SuspiciousButExpected
#7	HVMC - ePHI storage keys retrieved or storage exposure changed	Medium	18:22:36	18:44:10	21.6	BenignPositive / SuspiciousButExpected
#8	HVMC - ePHI storage keys retrieved or storage exposure changed	Medium	18:27:35	18:44:06	16.5	BenignPositive / SuspiciousButExpected

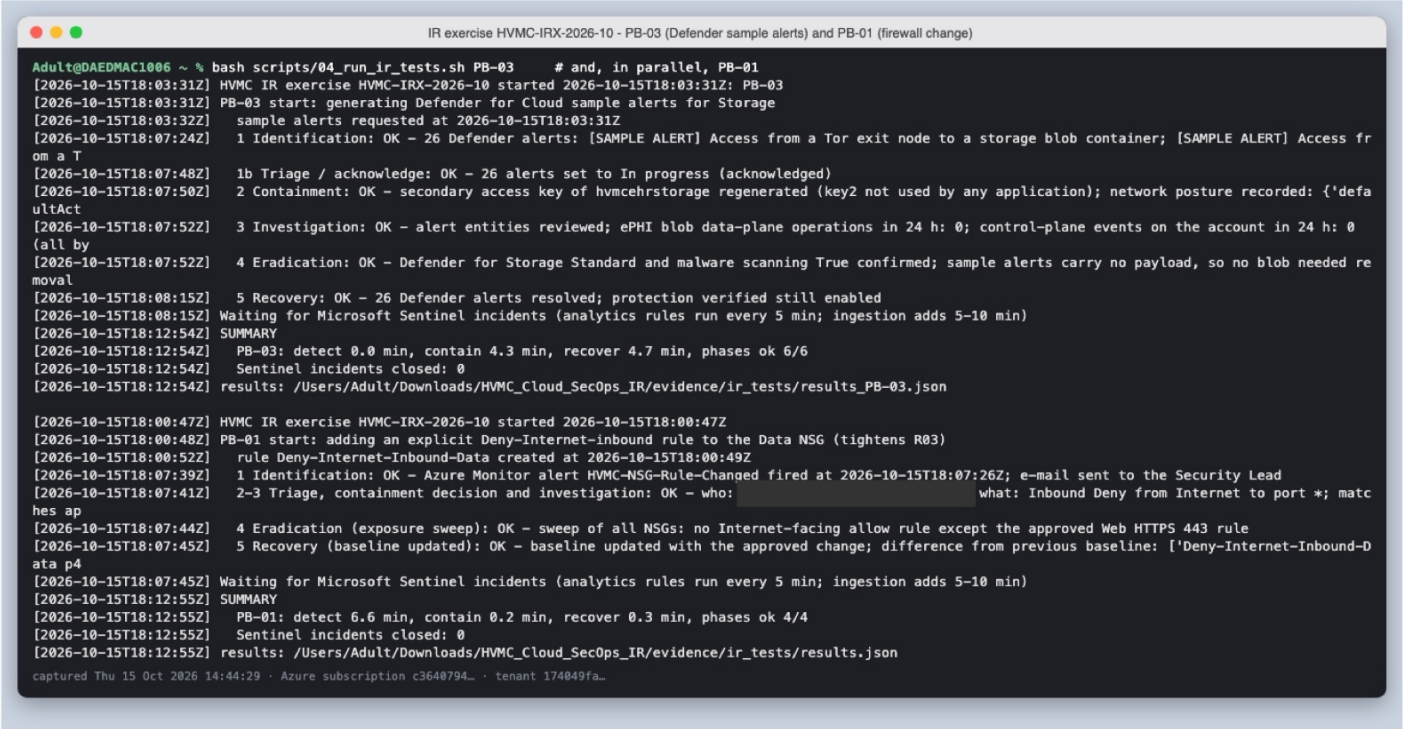


Figure 23. Live test run log: every playbook phase timestamped.

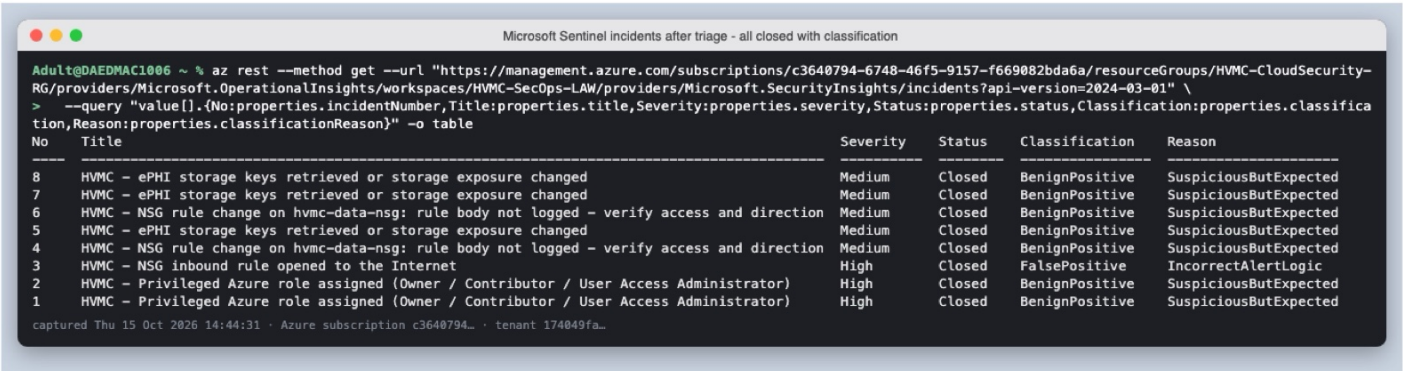


Figure 24. Sentinel incidents after the exercise: commented and closed with classification.

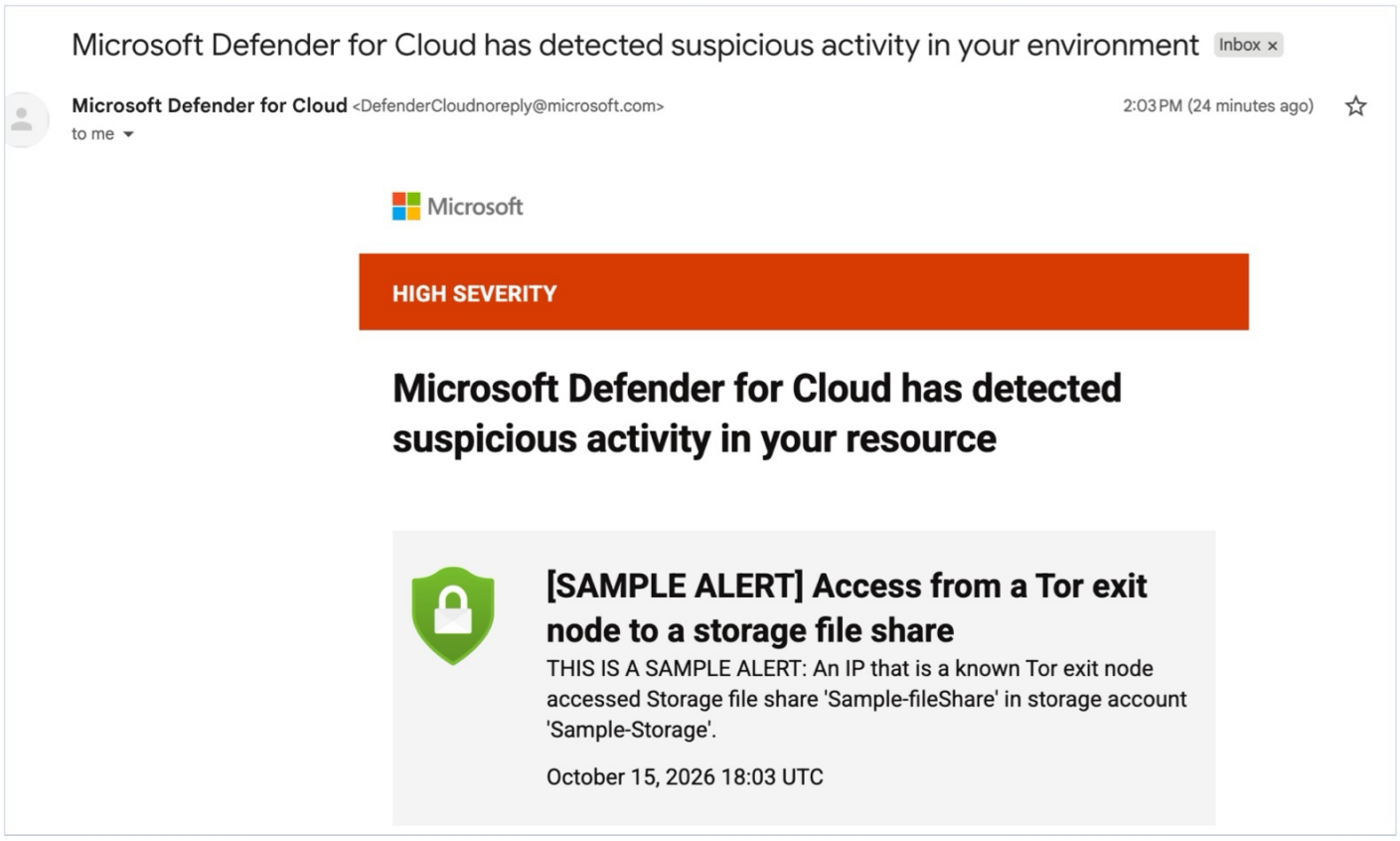


Figure 25. PB-03 identification: Defender for Cloud e-mail to the Security Lead - High severity sample alert, 18:03 UTC.

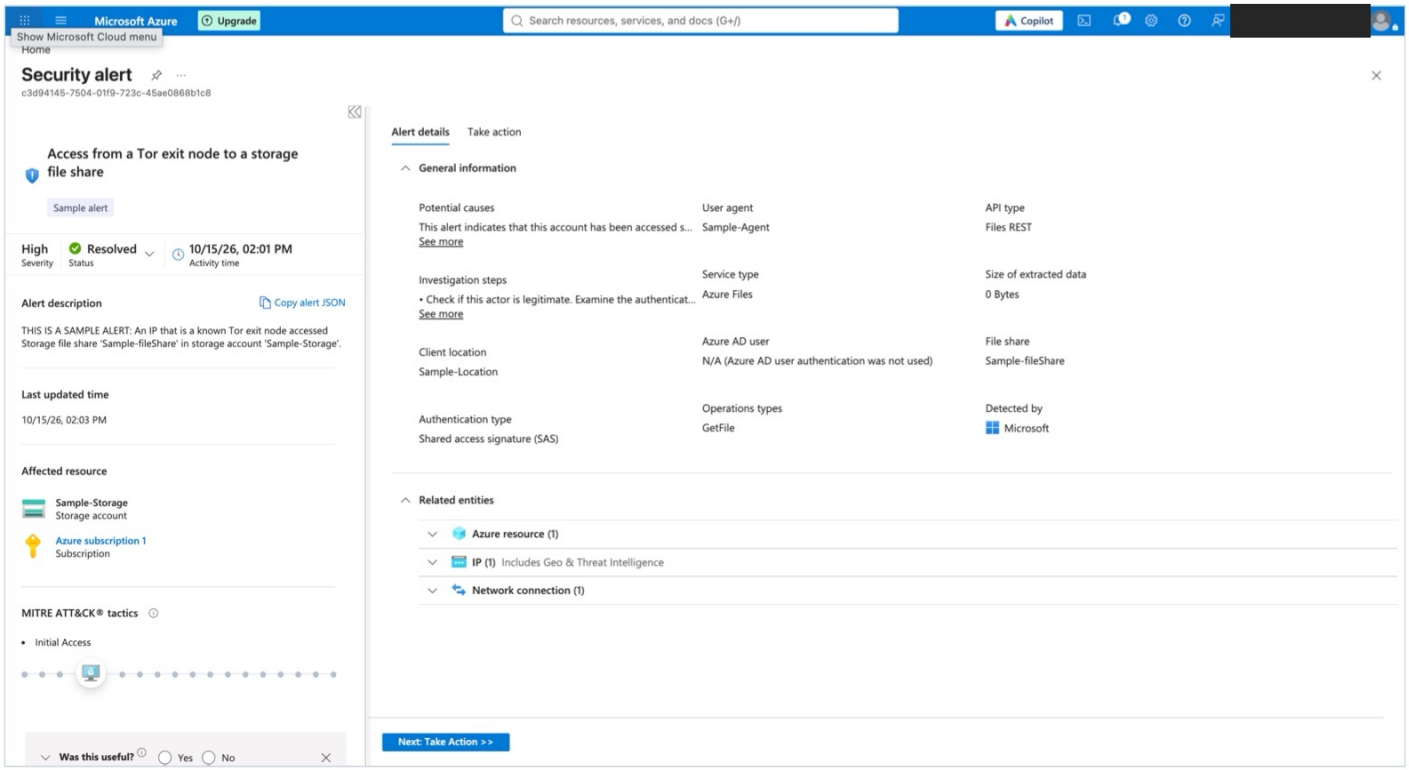


Figure 26. PB-03 recovery: the same alert in the Azure portal, status Resolved, with MITRE tactic, entities and investigation steps.

5.2 Tabletop exercise — PB-02

Run on real data: the eight HVMC-Role-Assignment-Added alerts of 17:12 UTC were the inject. The team identified them as platform-managed identities in 6 minutes, then rehearsed a variant (User Access Administrator granted to the

clinical test account) through containment, investigation, eradication and recovery. Findings and the improvement plan are in sections 7–8 of the playbook.